The Internal Data Commons

Admission, Producer Incentives, and the Financing of Rival Support

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Abstract

Reusable data are nonrival, but the support needed to apply them is scarce. We solve a model of endogenous producer investment, heterogeneous downstream applications, rival support and private producer costs. Support-weighted marginal-cost and scarcity charges implement efficient admission conditional on investment, but need not finance provision. An incremental-surplus score induces investment, and a restricted common-fee accounting class yields an exact participation and reserve frontier. A shared fee can prevent full-information implementation even when type-specific fees could finance it; the largest feasible credit share maximizes welfare within the class. The results retain the constrained baseline in payments and rents. All examples are synthetic, and no unrestricted mechanism optimum or firm-level effect is claimed.

Keywords: internal information; support capacity; admission; provision incentives; internal accounting.

1. Introduction

A dataset can be copied without being depleted. The same cannot be said of the people who explain definitions, resolve defects, approve appropriate use, and help downstream units apply the data. The distinction creates an allocation problem inside an organization. Technical facilitation can increase applications and their private benefits while generating an unpriced burden on a shared support function. At the same time, a producer may receive too little benefit from improving information that several other units reuse. A policy that addresses only one margin can worsen the other.

We solve both margins in a tractable setting. A producer improves reusable coverage or quality, denoted q, and documentation, denoted d. Both improvements benefit multiple use classes without being consumed. Consumer applications are heterogeneous in value and support requirements. They choose their use intensities after observing the producer's investments. Support has both a convex real cost and a hard capacity limit. Producer costs have a privately known type. Headquarters can audit selected economic primitives and operate a defined settlement rule; it does not observe the producer's cost type or set arbitrary contingent contracts.

The principal result is a necessary-and-sufficient financing condition for a transparent policy class. Efficient support admission and appropriate investment incentives require different entries in the ledger. Consumer charges reflect marginal support cost and scarcity. Producer payments reflect incremental downstream surplus after support costs, relative to the same organization's no-improvement baseline. A common producer entry fee can recover some of these payments without changing marginal incentives, but voluntary participation limits the fee by the lowest producer rent across cost types. The central reserve must cover the largest remaining settlement deficit. The type generating that deficit need not be the type that limits the entry fee.

This cross-type financing constraint is not solved by setting an appropriation share to one. Full appropriation implements the full-information allocation only when its payments are financeable. Otherwise, the strongest feasible common credit share is optimal within the accounting class. We derive this frontier and solve an example with two consumer classes and two distinct production margins. The result depends on the restriction to a common fee, audited surplus scoring, and no fixed consumer assessments. We do not claim it is an unrestricted optimal mechanism or a new general impossibility theorem.

Rasouli and Jordan (2021) is the closest data-sharing mechanism comparator. Our additional object is the joint choice of costly improvements and congestible applications, together with the exact reserve needed when one common fee must satisfy every producer type. This is a conditional accounting frontier, not a general financing impossibility or a new truthful-reporting mechanism.

A second result concerns the investment score itself. A seemingly attractive reward for investment value multiplied by actual usage has a strategic defect. Once investment changes downstream demand, the producer can expand its reward through the induced change in usage. The correct surplus score uses an envelope property and does not pay that extra derivative. This distinction matters even with honest, verified use counts: gaming is not needed for the distortion.

The former manuscript's quantity-versus-surplus distinction remains a benchmark. It is no longer the article's main contribution. An exact two-margin example shows that a technical intervention can reduce the incentive interaction in q while increasing its interaction in total surplus, with adoption endogenous. Documentation expands in that example. No scalar measure of provision captures every margin.

The numerical work consists of analytical examples and reproducibility checks, not calibration or evidence that a particular firm suffers these distortions. A future experiment can investigate their importance. Its prospective protocol and its toy exposure-estimator check are separate from the completed theoretical contribution.

2. Literature and the contribution boundary

Jones and Tonetti (2020) establish the economic importance of nonrival data and analyze property rights, interfirm hoarding, creative destruction, and privacy. Our internal application adds rival support and decentralized budgets; relocating nonrivalry inside a firm is not itself a new insight. Cabrera and Cabrera (2002) already describe internal knowledge-sharing dilemmas, and Siemsen, Balasubramanian, and Roth (2007) analyze incentives for effort, helping, and knowledge sharing.

The distinction between nonrival provision and congestible use belongs to club and local-public-goods theory. Buchanan (1965) studies the preferred sharing group. Berglas and Pines (1981) explicitly study optimal pricing, provision, and conditions for financial surplus or deficit. Consequently, neither congestion pricing nor the observation that efficient user prices may fail to finance provision is claimed as new here. Their published summary was verified; a full reprint was not obtained in this run.

Rasouli and Jordan (2021, version 2, Sections 4-5) is the closest data-sharing mechanism. Proposition 4.1 implements every welfare-maximizing sharing graph with competitive prices under additive pairwise sharing costs and unrestricted replication. Section 5 instead has private utility/cost types, a planner intermediary, and monetary transfers plus individually targeted quality distortion. Its mixed-VCG construction assumes zero-cost upward/downward distortion with isolated effects on other agents; the distorted variant starts from degraded quality. These are substantive implementation instruments, not provided here. Our improvements are costly and chosen before use. Convex aggregate support and hard capacity couple recipients, violating the additive unrestricted-sharing structure. Our audited schedule is not a private-reporting solution. The incremental result is the exact cross-type fee/reserve constraint (23), where one type may limit the fee and another require the largest settlement, and its largest-feasible-share allocation. This frontier applies only to our fixed score, common fee, and absence of consumer fixed levies. It neither rules out their instruments nor improves their VCG claims. Their version also distinguishes agent welfare from total welfare; its equation (31) retains a budget-imbalance term, so we do not infer an unconditional undistorted-welfare guarantee from its proposition

wording.

Bierbrauer (2011), whose institutional working version we examined, is a close financing comparator. Its regulator approves a supplier-proposed public-good mechanism under alternative consumer-protection rules; cost uncertainty affects the ability to extract producer rents. Our narrower exercise fixes a support-pricing and surplus-score family, preserves real congestion, and derives its exact ex-post reserve requirement and common-fee constraint. We do not improve its general theory of incomplete contracting or justify a self-financing requirement universally.

Dikolli and Vaysman (2006) distinguish coarse reports, operational incentives, and internal trade. Their cost-based rule uses the production budget, divisional profit shares guide operations, and fixed salaries induce reporting; negotiated pricing instead uses offers and responses with costly delay. Their comparison turns on communication and bargaining. Here the allocation is repeated reuse with support congestion; the score and common fee solve a different restricted financing problem. The accessible article-in-press reprint's Sections 2-6 were examined, including the revenue-based qualification to their comparison. The official publisher record confirms DOI 10.1016/j.jacceco.2005.06.001 and final pages 201-234.

Jamm (2018) separates internal knowledge creation, sharing, and application and adds a knowledge-management system that changes costs and contracting information. d'Aspremont, Bhattacharya, and Gerard-Varet (1998) provide richer private-information implementation results for shared knowledge and development effort. Our results do not supplant either analysis. They establish a particular admission-investment-settlement solution, not the first model of internal information incentives. The strategic score example is also related to the established concerns about incentive measures emphasized by Holmstrom and Milgrom (1991).

The source ledger distinguishes full relevant-section reading, publisher summaries, and unsuccessful access attempts. The article offers a precisely delimited derivation; a comprehensive priority claim across all congestion, procurement, and public-goods models is not established. Its strongest remaining intellectual risk is that the restricted financing frontier may be judged a modest specialization of those literatures.

3. Environment and information

3.1 Nonrival improvements and rival applications

An organization already holds a source used operationally. The producer selects two nonnegative improvements $z = (q, d)'$. The first is maintained reusable coverage or quality; the second is maintained documentation. They are distinct investments with distinct downstream productivities. The source and its improvements are nonrival conditional on support. This is not a model in which increasing q manufactures private copies of data.

There are $J \geq 2$ eligible downstream application classes. Their masses of realized use are $y_j \geq 0$. The class's gross gain net of its own implementation effort is

$$u_j(y_j, z; t) = (h_j(t) + v_j' z)y_j - \frac{y_j^2}{2n_j}, \qquad n_j > 0. \tag{1}$$

Here v_j is row j of $V \geq 0$, and h_j is baseline value net of integration cost. A rise in h_j can represent an adapter that lowers integration costs while leaving underlying accuracy unchanged. The quadratic term can equivalently be derived from a continuum of atomistic application opportunities with increasing marginal implementation cost. Individual opportunity costs need not be reported to headquarters. In numerical examples V has full column rank: the two investments are not a frozen scalar recipe.

Legitimate access restrictions are applied first. An ineligible class is constrained to $y_j = 0$ and removed from the eligible system. Neither prices nor rewards can override that constraint.

The economic admission problem concerns eligible uses only. Positive usage does not define data quality, decision rights, or successful decision making.

An application in class j consumes $a_j > 0$ support units. Total support load and real support cost are

$$\ell = a'y, \qquad S(\ell) = \frac{\gamma}{2}\ell^2, \quad \ell \le \bar{L}, \quad \gamma \ge 0, \quad \bar{L} > 0. \tag{2}$$

The hard limit is an existing staffing or service-capacity constraint. Capacity expansion is not a choice in this article. Producer support can be administered by a shared desk or through a separately audited producer service account. In either case its real cost enters (2) once. Separate accounting prevents cost reimbursement from being mistaken for a reward for data quality.

Producer investment cost is

$$C_\kappa(z) = \frac{\kappa}{2} z'Kz, \quad K = \operatorname{diag}(k_q, k_d) \succ 0, \quad \kappa \in \mathcal{K} = \{\kappa_L, \kappa_H\}, \quad 0 < \kappa_L < \kappa_H. \tag{3}$$

Its type κ is privately known to the producer. More finite cost types can be included without changing the financing theorem. Put $D = \operatorname{diag}(1/n_j)$, $H = D + \gamma aa'$, and $Q = VK^{-1}V'$. We impose the transparent sufficient condition

$$D - \frac{1}{\kappa_L} Q \succ 0. \tag{A}$$

It ensures strict joint concavity of the relevant allocation programs, including the unpriced comparison, for credit shares in $[0, 1]$. It is sufficient rather than necessary. Technologies with strong increasing returns that violate (A) are outside the results.

Organizational surplus before a policy's real administration cost is

$$W_\kappa(z, y; t) = h'y + z'V'y - \frac{1}{2} y'Dy - \frac{\gamma}{2}(a'y)^2 - C_\kappa(z). \tag{4}$$

The first two terms are gains relative to task-level alternatives; overlap must already be removed from their measurement. Internal payments, entry fees, and budget transfers do not enter (4). An incremental accounting and audit cost $m \ge 0$, constant across the policies in our accounting class, is subtracted once from surplus and paid once in the settlement budget. Allowing m to depend arbitrarily on incentives would change the monotonicity result below.

3.2 Observable, auditable, and private quantities

Outputs q, d, eligible application identities, and support loads are observable and auditable. The exact scoring analysis additionally requires an enforceable aggregate value schedule (1), the productivities V, and the support technology. This is a substantive auditing assumption. It can describe a restricted set of tasks with measured implementation effort and validated loss reductions; it does not describe every corporate dataset. Accounting for time released as a benefit requires usable capacity or another explicit shadow-value convention.

The producer's opportunity cost type κ is not verified. We do not solicit a cost report or assume a truthful cost budget. Consumers may have private individual opportunity costs, but the aggregate inverse-demand schedule must be validated to compute the investment score. If h, D, V are not auditable, exact implementation is unavailable. Section 7 analyzes a bounded scoring error rather than substituting an unverified value claim for this assumption.

Headquarters can levy support prices and a common producer participation fee, pay a verified-surplus score, and supply a limited central reserve $B \ge 0$. It cannot levy fixed consumer assessments or select a different fixed producer fee after observing the private cost type. These

are explicit restrictions of the studied accounting class, not universal technological impossibilities. Implementation means the producer's optimal sequential investment and consumers' optimal price-taking use under the announced ledger; no dominant-strategy reporting theorem is claimed. Nonlinear screening menus, bargaining, and output-contingent rent extraction are excluded. Observed outputs might reveal a type ex post, but converting that inference into a contingent fee would change investment incentives and the contract class.

4. Unpriced openness and the full-information allocation

4.1 A coherent decentralized comparison

Let $\bar{y}_j$ be nontransferable support reservations satisfying $\sum_j a_j \bar{y}_j \leq \bar{L}$. Consumers face no support charge and choose

$$y_j^D(z) = \min\{\bar{y}_j, n_j[h_j + v_j' z]_+\}. \tag{5}$$

Define $G_D(z) = \max_{0 \leq y \leq \bar{y}}\{h'y + z'V'y - y'Dy/2\}$. A producer receiving a share $\alpha \in [0,1]$ of verified incremental consumer surplus chooses

$$z_D \in \arg\max_{z \geq 0}\{\alpha[G_D(z) - G_D(0)] - C_\kappa(z)\}. \tag{6}$$

This is a sequential equilibrium: investment precedes use and the producer anticipates adoption. Its score omits the separately funded support burden. Thus the comparison need not rely on raw download rewards or dishonest accounting. It illustrates an internal boundary that attributes consumer gains while leaving support costs outside the incentive score. Administration and financing of this status quo are not assumed identical to the corrective mechanism's reserve constraint.

Proposition 1 (unpriced equilibrium and harmful opening). Under (A), (5)-(6) have a unique allocation. On a branch with all uses strictly between their reservation bounds,

$$y_D = (D - \alpha Q/\kappa)^{-1} h, \qquad z_D = \frac{\alpha}{\kappa} K^{-1} V' y_D. \tag{7}$$

Its actual surplus is

$$W_D = \frac{1}{2} y_D'[D - \alpha^2 Q/\kappa - \gamma a a']y_D. \tag{8}$$

A marginal increase in h_j lowers actual surplus precisely when

$$e_j'(D - \alpha Q/\kappa)^{-1}[D - \alpha^2 Q/\kappa - \gamma a a']y_D < 0. \tag{9}$$

This can occur with two full-rank production margins, positive consumer gains, and slack hard capacity. It need not occur in every firm. The result isolates an unpriced support externality; the private-use costs in (1) are already internalized.

4.2 Full information and conditional admission

With known κ, the resource benchmark maximizes (4) over $z, y \geq 0$ and $a'y \leq \bar{L}$. It is unique under (A). Its first-order conditions are

$$\kappa K z_F = V' y_F, \qquad 0 \leq y_{F,j} \;\perp\; (Dy_F)_j + a_j(\gamma \ell_F + \nu_F) - h_j - v_j' z_F \geq 0, \tag{10}$$

$$\nu_F \geq 0, \quad a'y_F \leq \bar{L}, \quad \nu_F(a'y_F - \bar{L}) = 0. \tag{11}$$

The producer condition holds componentwise with complementary slackness; nonnegative V and diagonal K make the displayed equality valid even for zero components. A zero-productivity investment is zero.

Conditional on any z, charge each consumer a price per application

$$p_j = a_j\lambda, \quad \lambda = \gamma\ell + \nu, \quad \nu \geq 0, \quad \nu(\ell - \bar{L}) = 0. \tag{12}$$

Atomistic users then select the unique support-efficient use allocation. The market or admission desk adjusts the support-unit price until demand respects capacity. No report of producer cost is required. Some eligible classes receive zero use when their marginal value does not cover support and scarcity. Conversely, unused support capacity does not imply every positive-private-value use should be admitted: convex support cost can justify exclusion before the hard cap binds.

Proposition 2 (admission and the wrong pricing unit). Rule (12) implements the full-information use choice conditional on investment. If two classes are interior, $a_1 \neq a_2$, and $\lambda > 0$, a single identical per-application price cannot implement the same allocation. The required prices are $a_1\lambda$ and $a_2\lambda$. A price on copied bytes or raw uses therefore cannot generally stand in for a price on rival support.

This is an application of marginal-cost and scarcity pricing, not a new congestion-pricing theorem. Its role is to specify what the corrective ledger must measure before provision rewards are considered.

5. Strategic investment and a feasible surplus score

Define the support-efficient downstream value at investment z by

$$G(z;t) = \max_{y \geq 0,\ a'y \leq \bar{L}} \{h'y + z'V'y - \tfrac{1}{2}y'Hy\}, \qquad G_0(t) = G(0;t). \tag{13}$$

An announced credit share $\beta \in [0,1]$ pays

$$T_\beta(z;t) = \beta[G(z;t) - G_0(t)]. \tag{14}$$

The producer pays a fixed fee $F \geq 0$ upon joining and chooses investment to maximize $T_\beta(z) - C_\kappa(z) - F$, anticipating subsequent use and support prices. The scoring schedule is committed ex ante. The benchmark G_0 uses the same demand and technical environment with zero producer improvements; it is not last year's outcome under different demand. The center pays for verified incremental surplus, not total existing data value.

This timing avoids treating a monopolistic producer as a price taker in a usage-dependent credit. First headquarters announces the score, support rule, and fee; the producer learns its cost and accepts or declines; it chooses auditable improvements; consumers adopt at (12); and the ledger settles. The producer anticipates its effect on adoption. Auditing the aggregate schedule in (13) is what permits the envelope-based score to handle this effect.

Proposition 3 (joint allocation under a surplus score). Conditional on participation, for $\beta > 0$ the unique allocation solves

$$\max_{z,y \geq 0,\ a'y \leq \bar{L}} \left\{ h'y + z'V'y - \tfrac{1}{2}y'Hy - \frac{\kappa}{2\beta}z'Kz \right\}. \tag{15}$$

In particular,

$$z_\beta = \frac{\beta}{\kappa} K^{-1} V' y_\beta. \tag{16}$$

On any positive-use set S, let $M_\beta = H - \beta Q/\kappa$. Then

$$y_{\beta,S} = M_{\beta,SS}^{-1}(h_S - \nu_\beta a_S), \quad y_{\beta,S^c} = 0, \tag{17}$$

where $\nu_\beta = 0$ if capacity is slack, and otherwise

$$\nu_\beta = \frac{a_S' M_{\beta,SS}^{-1} h_S - \bar{L}}{a_S' M_{\beta,SS}^{-1} a_S}. \tag{18}$$

Only sets satisfying nonnegativity, the inactive-class inequalities, and capacity complementarity are valid. These formulas solve admission and both investment margins jointly. At $\beta = 0$, $z = 0$ and y maximizes (13) at zero improvement. At $\beta = 1$ the allocation is the full-information resource benchmark, conditional on participation and financeability, not yet a feasible-budget policy.

Proposition 4 (score levels and why gross-use credits are different). On a region with positive use and slack capacity, set $b = V'H^{-1}h$, $J = V'H^{-1}V$, and $c_0 = \frac{1}{2}h'H^{-1}h - G(0)$, with $G(0)$ the actual constrained value in (13). The correct surplus score on this region equals

$$T_\beta(z) = \beta(c_0 + b'z + \tfrac{1}{2}z'Jz). \tag{19}$$

The constant is zero if zero investment is on the same full positive-use, slack-capacity branch (or its closure with the same value). Otherwise it must be retained. The unconstrained quadratic value at zero cannot replace $G(0)$. A branch-specific constant leaves the investment derivative on that branch unchanged, but affects payment levels, participation rents and fixed-fee budget constraints. Formula (19) applies only on its stated branch; the global definition remains (14).

A gross-use reward $\beta z'V'y(z)$ instead equals $\beta(b'z + z'Jz)$. For $\beta > 0$, whenever $\kappa K - 2\beta J \succ 0$ and both candidate investments and their uses satisfy the nonnegativity and slack-capacity branch conditions, the branch stationary solutions are

$$z_{\text{gross}} = (\kappa K - 2\beta J)^{-1}\beta b, \quad z_\beta = (\kappa K - \beta J)^{-1}\beta b. \tag{20}$$

At $\beta = 1$, if $Jz_F \neq 0$ and the gross optimum remains on this branch, it differs from the unique resource optimum and strictly lowers surplus. It pays an extra derivative for demand that investment itself induces. Honest verified uses alone therefore do not make a usage-multiplied reward an efficient provision incentive.

6. Participation, the central reserve, and constrained optimal policy

Let $c_\kappa(\beta) = C_\kappa(z_\beta)$ and define the producer's optimized rent before the common fee by

$$r_\kappa(\beta) = T_\beta(z_\beta) - c_\kappa(\beta) \geq 0. \tag{21}$$

The inequality follows because $z = 0$ yields zero incremental score and zero cost. Support prices generate revenue $\lambda\ell$. Net of real support expenditure, the support account has surplus

$$\Gamma_\kappa(\beta) = \lambda_\kappa \ell_\kappa - S(\ell_\kappa) = \frac{\gamma}{2}\ell_\kappa^2 + \nu_\kappa \ell_\kappa. \tag{22}$$

The accounting center receives support revenue and the producer fee; pays support resources, the score, and administration; and may draw up to B from its central reserve. Producer participation is voluntary after its type is known. Its outside incremental payoff is zero. Both participation and the reserve limit are required ex post, for every $\kappa \in \mathcal{K}$. A weakly indifferent producer participates by the specified tie rule.

Theorem 1 (exact common-fee reserve requirement). A given β is implementable in this accounting class if and only if

$$B \geq B_I(\beta) := \left[m + \max_{\kappa \in \mathcal{K}} \{T_\kappa(\beta) - \Gamma_\kappa(\beta)\} - \min_{\kappa \in \mathcal{K}} r_\kappa(\beta) \right]_+ . \tag{23}$$

Here $T_\kappa(\beta) = T_\beta(z_\beta(\kappa))$. An optimal common fee for minimizing the reserve is $F = \min_\kappa r_\kappa(\beta)$. All feasible fees satisfy

$$\max\{0, m + \max_\kappa (T_\kappa - \Gamma_\kappa) - B\} \leq F \leq \min_\kappa r_\kappa. \tag{24}$$

If producer cost were verifiable and type-specific fixed fees allowed, the corresponding reserve would be

$$B_K(\beta) = \left[m + \max_\kappa \{c_\kappa(\beta) - \Gamma_\kappa(\beta)\} \right]_+ \leq B_I(\beta). \tag{25}$$

The inequality can be strict. The private-cost/common-fee constraint then blocks an otherwise financeable allocation. Equations (23)-(25) are specific to this accounting class. They are not a mechanism-independent lower bound, a claim that all producer rents are unavoidable, or a new general private-information impossibility.

The structure is informative even when internal transfers cancel from welfare. Marginal support charges finance rival service and may generate congestion or scarcity rents. Those rents need not cover nonrival producer incentives. A common fixed fee cannot be made large enough to fund an expensive settlement state if doing so causes a different cost type to decline. The gap arises from jointly respecting the admission rule, the provision score, and voluntary participation.

Theorem 2 (constrained allocation frontier). Actual resource surplus $W_\kappa(z_\beta, y_\beta)$ is weakly increasing in $\beta \in [0, 1]$ for every type. If

$$\mathcal{B}(B) = \{\beta \in [0, 1] : B_I(\beta) \leq B\} \tag{26}$$

is nonempty, its largest element $\beta^*(B)$ exists and maximizes organizational surplus minus m within the accounting class, separately for every type and therefore for any distribution over types. The proof requires neither an interval assumption for the feasible set nor monotonic reserve needs. We do not establish that disconnected feasible sets occur within this maintained model. If (26) is empty, this accounting system cannot be operated with the stated reserve and administration cost.

At $\beta = 1$, full information is achieved only if (23) holds. Under a binding reserve, $\beta^* < 1$ is a solved constrained policy, not an unimplemented recommendation to internalize all benefits. Changing support capacity, permitting consumer fixed fees, or allowing screening menus may move the frontier; these are different governance regimes.

Weak participation is relevant at the frontier. If every type must strictly prefer joining, replace the upper fee bound by $F < \min r$. A weakly feasible boundary with no slack is not attained under strict participation. It is a strict-participation supremum only if strictly feasible shares approach it; that approximation is not guaranteed. If $\min r > 0$, a small reserve increase, or a lower share with financing slack, can support a positive participation margin by reducing the fee

below every rent. If $\min r = 0$, strict participation is impossible with $F \geq 0$ at that share; extra reserve alone does not fix it. This qualification is not hidden in the numerical policy.

7. Endogenous use, multiple margins, and imperfect verification

7.1 What survives the old comparative static

With the adoption set fixed, the former scalar benchmark had $q_\alpha = B/K_c$ and induced-surplus derivative $(1-\alpha)B^2/K_c$. It generated different proportional-growth thresholds for quantity and surplus complementarity. Appendix F preserves the proof as a benchmark only.

Here investment and use both respond. On a smooth branch, the reduced investment problem has first-order condition $\beta \nabla G(z) = \kappa K z$. Differentiating gives

$$z_\beta = [\kappa K - \beta G_{zz}]^{-1} \nabla G, \quad \frac{dW_\kappa}{d\beta} = \frac{1-\beta}{\beta} \kappa z' K z_\beta. \tag{27}$$

There is no common scalar denominator or single B governing all investment quantities. Cross-technical derivatives depend on shifts in use composition, both investment margins, and support scarcity. Theorem 2 establishes monotonic welfare in the credit share; it does not sign every technical-incentive interaction.

Nevertheless, opposite quantity and surplus interactions can survive endogenous use. An exact full-rank example in Section 8 has a negative factorial interaction in q, a positive interaction in d, and a positive interaction in welfare. This is an existence result on a declared additive scale. Adding q and d in arbitrary units would change the quantity statistic and is not justified.

7.2 An explicit audit limit

Suppose the entire incremental score is overstated by a common factor $1+\delta$, with $0 \leq \delta \leq \bar{\delta}$, while actual consumer behavior and support pricing remain as modeled. The producer then responds to an effective share $\widetilde{\beta} = \beta(1+\delta)$. Choosing $\beta \leq 1/(1+\bar{\delta})$ ensures that effective incentives do not exceed the efficient endpoint and that resource surplus is at least the zero-credit support-priced benchmark, before any change in administration cost. The reserve calculation must use the actual inflated payments and worst-case rent constraints, not the uninflated formula with an assumed truthful score.

If $\widetilde{\beta} > 1$, overinvestment can lower welfare. Proposition 4 gives a distinct example where the wrong scoring functional form creates the same issue without lying. If no reliable bound on counterfactual value is auditable, the positive-share guarantee is unavailable. Fraudulent duplicate tasks, collusion between producers and consumers, private support intensity, and manipulation of the baseline are not solved by this model. Exact implementation is conditional on the measured object being enforceable.

8. Solved examples and computational verification

All examples are synthetic resource units. They are constructed to exhibit analytical mechanisms and are not estimates of an organization's losses. No additional parameter sweep is offered as evidence of novelty.

8.1 Opening can harm while provision increases

Use $K = \mathrm{diag}(4,5)$, $V = ((.5,.1),(.1,.4))$, $n = (1,1)$, $a = (1,1)$, $\gamma = .5$, $\bar{L} = 20$, $\kappa = 1$, and $\alpha = .5$. Both producer margins are independent: $\det V = .19 > 0$. Technical facilitation raises h_2 from -1 to $.5$ while $h_1 = 4$. Reservation limits are $(10,10)$.

Accounting rule	h_2	q	d	y_1	y_2	Surplus
Unpriced support	-1.0	.2583	.0413	4.1333	.0000	4.1333
Unpriced support	.5	.2656	.0635	4.1391	.5519	3.0658
Support priced, $\beta = .5$	-1.0	.1703	.0273	2.7253	.0000	5.5104
Support priced, $\beta = .5$	.5	.1703	.0273	2.7253	.0000	5.5104

The new class's private gain is positive, both investments increase, and hard capacity remains slack. Yet unpriced surplus falls by 1.0675 because the new support burden is not included in consumer adoption or the producer score. The corrective admission price excludes this low-value class and prevents this loss in the example. This is not a claim that efficient admission universally requires rejecting newly feasible uses.

8.2 Private cost and a constrained zero-reserve policy

Set $h = (2, 1)$, $K = 2I$, $V = ((1, .2), (.2, .8))$, $n = a = (1, 1)$, $\gamma = .15$, $\bar{L} = 10$, $\mathcal{K} = \{1, 2\}$, $m = 0$, and $B = 0$. The decentralized comparison uses $\alpha = .2$ and reservations $(5, 5)$. Table 2 compares its allocation with support pricing alone, the optimal implementable share, and the resource benchmark.

Cost type	Regime	Credit share	q	d	Surplus
1	Decentralized	.2000	.2511	.1385	2.3008
1	Support price only	.0000	.0000	.0000	1.9808
1	Constrained implementable	.5371	.6507	.3159	3.1473
1	Full information	1.0000	1.7550	.8657	3.9164
2	Decentralized	.2000	.1173	.0643	2.0402
2	Support price only	.0000	.0000	.0000	1.9808
2	Constrained implementable	.5371	.2760	.1329	2.4455
2	Full information	1.0000	.5912	.2867	2.6307

The largest feasible share is $\beta^* = .5370860743$, with common fee $F = .1627317858$. At cost type 1 the center's budget binds; type 2 sets the participation bound. The type-1 rent before the fee is .3843, while the type-2 rent is .1627. Type 2 weakly prefers participation under the stated tie rule. The benchmark at $\beta = 1$ requires reserve 3.5326 with a common fee, versus 2.2469 with verifiable cost and type-specific fees. Thus even reserve 2.2469 would fund each known-type efficient allocation but cannot fund both under the common fee. Administration is normalized to zero to isolate this wedge; a real positive m enters (23) and reduces net welfare.

A separate binding-capacity solution uses the same primitives but $\bar{L} = 2$. At $\beta = 1$, cost type 1 has $(y_1, y_2) = (1.7867, .2133)$ and scarcity price $\nu = .8808$; type 2 has $(1.6229, .3771)$ and $\nu = .5330$. Both meet capacity exactly. The model therefore includes genuine capacity rationing, rather than substituting a convex-cost example for the hard constraint.

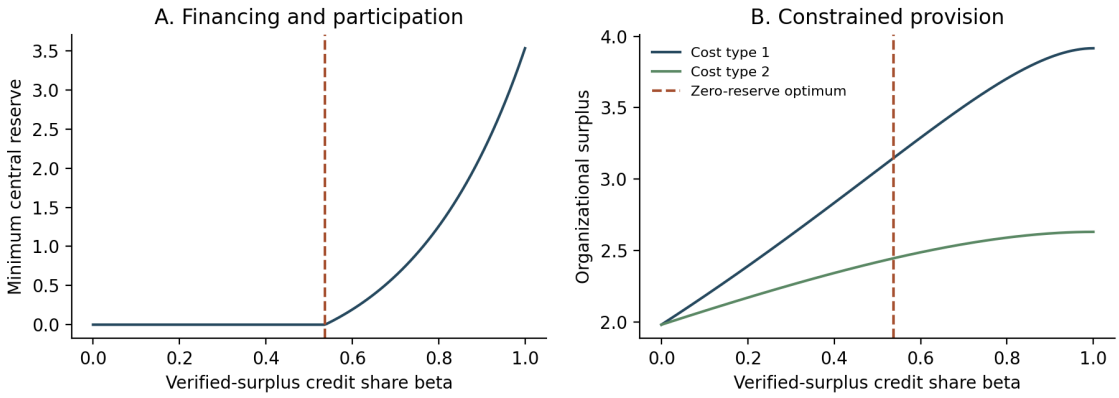


Figure 1: Synthetic example: the zero-reserve frontier and surplus under both private cost types. The dashed line marks the largest feasible common credit share.

8.3 Opposite interactions with two production margins

Use $K = 2I$, $V = ((1, .1), (.1, 1))$, $n = (1, 1)$, $a = (1, 2)$, $\gamma = .1$, $\bar{L} = 100$, and $\kappa = 1$. Facilitation changes $h = (4, 1)$ to $(4, 2)$. Compare credit shares .4 and .7 under the support-priced surplus score. All classes are interior and $\det V = .99$. The exact factorial differences are

$$\Delta q = -.0119774850, \qquad \Delta d = .1679228408, \qquad \Delta W = .1266561870. \tag{28}$$

Rational matrices and exact fractions in the reproduction output certify the signs. This shows survival of the sign distinction as an allocation example, not survival of the old scalar growth formula. The policies in this example need not be financeable with $B = 0$; financing is separately governed by (23).

8.4 Verification actually completed

The active-set solver enumerates positive-use sets and capacity states. An independent optimization of the original four-variable program checks eight declared cases: interior use, excluded classes, no demand, zero credit, and binding capacity. Maximum objective discrepancy was 2.87×10^{-13} in the completed run. Exact rational calculations isolate all financing-inequality roots in $[0, 1]$ for Table 2 and verify the largest feasible boundary, rather than assuming the reserve function is monotone. Exact rational contrasts check (28). A strategic gross-credit example independently verifies the strict welfare loss in Proposition 4.

These calculations validate implementations of stated equations. The proofs establish the results. Neither establishes empirical prevalence, an effect size in euros, or the correctness of a firm-specific value audit. The separately retained toy exposure estimator is not evidence for the organizational mechanism and is not part of the theoretical verification.

8.5 Exact regression across the baseline branch

This single synthetic regression checks score levels and settlement, not a parameter sweep. Set $D = I$, $a = (1, 1)'$, $K = 2I$, $V = ((1, 1/5), (1/5, 4/5))$, $h = (4, -1/10)'$, $\gamma = 3/20$, $\bar{L} = 10$, and $\kappa = \beta = 1$. Rational active-set enumeration, separately implemented from the floating-point solver, gives

$$y_0 = (80/23, 0)', \qquad G(0) = 160/23.$$

The first baseline KKT equation is $(23/20)y_{0,1} = 4$, while the excluded class has marginal gain $-143/230 < 0$. The unconstrained full-use baseline $H^{-1}h = (71/20, -11/20)'$ is infeasible and has quadratic value 2851/400. Hence

$$c_0 = \frac{2851}{400} - \frac{160}{23} = \frac{1573}{9200} > 0. \tag{29}$$

Solving $(H - VK^{-1}V')y = h$ and $z = K^{-1}V'y$ independently gives

$$y = (5395/849, 95/849)', \qquad z = (2707/849, 385/566)'.$$

Both uses are positive, $a'y = 1830/283 < 10$, and $\nu = 0$. Condition (A) and all investment/use KKT equalities are checked exactly. Strict concavity therefore makes this the unique global surplus-score allocation. At this allocation $G(z) = 33636065/1441602$. For the incomplete R1 branch polynomial $P = b'z + z'Jz/2$,

$$P - [G(z) - G(0)] = -1573/9200.$$

The exact ledger below evaluates the incomplete level at the same allocation; it does not claim that the incomplete expression defines a valid global scoring rule.

Quantity	Incomplete branch polynomial	Actual constrained-baseline score
Producer payment T	4672209349/288320400	542973175/33156846
Rent $r = T - C_\kappa$	1893601/339600	448883/78108
Center deficit at fixed $F = 0$	3768006349/288320400	219494915/16578423
Maximum weak-IR fee, this single type	1893601/339600	448883/78108
Reserve at that maximum fee	21603391/2883204	21603391/2883204

Production cost is 30645421/2883204; support cost and support-account surplus are both 502335/160178; support revenue is 502335/80089. Actual payment is 16.3758994146, rent 5.7469529370, and fixed-zero-fee deficit 13.2397945812. Each exceeds its incomplete counterpart by $c_0 = .1709782609$. In this single-type ledger, raising the maximal weak-IR fee by the same amount cancels the reserve correction, giving 7.4928416442. At that fee participation is weak only. Reducing the fee by $\epsilon > 0$ gives strict participation and raises expenditure by ϵ.

In general, conditional on unchanged allocations, payment corrections Δ_κ change the reserve to

$$\left[m + \max_\kappa(T_\kappa + \Delta_\kappa - \Gamma_\kappa) - \min_\kappa(r_\kappa + \Delta_\kappa)\right]_+. \tag{30}$$

At fixed common fee, each state's expenditure changes by Δ_κ. A correction common to every type cancels from the maximal-fee reserve; unequal corrections can change the fee-limiting and deficit-generating types. Equal local derivatives alone guarantee neither unchanged global branch choice nor unchanged financing. The global score (14) is required.

Table 2 does not change: its baseline is $y_0 = (43/26, 17/26)'$, with both classes positive and slack capacity, so $c_0 = 0$. R1's allocation code already calculated the constrained $G(0)$. Rerunning prior examples and exact certificates reproduces $\beta^* = .5370860743$, $F = .1627317858$, $B_I(1) = 3.5326121937$, and $B_K(1) = 2.2469100348$. The regression compares the actual floating-point score, rent and support ledger to the exact values and checks preserved R1 numerical outputs. It rejects the incomplete score level even though a derivative-only test would pass.

9. Conclusion

The allocation lesson is to separate the accounting units. Charge for rival support and scarcity, and assess nonrival producer improvements against verified incremental surplus. Using an average byte price to recover all producer payments can distort heterogeneous admission. Conversely, a technically correct support price can leave provision underfunded. A common fee may recover rents without changing marginal decisions, but voluntary participation caps how much it can raise.

The reserve constraint is a distribution and commitment constraint inside an organization. It is not an additional social expenditure. A draw from the reserve or a payment to the producer is a transfer; actual investment effort, support, consumer implementation, and administration are the resource costs. Scarcity rent is not counted as an extra benefit in (4). The ledger can reconcile transfers without pretending that they disappear from unit-level incentives.

The result does not select a data architecture or vendor. A central authority with unrestricted consumer levies, fully verifiable producer costs, or sophisticated screening may achieve a better frontier. A value schedule that cannot be audited may make the proposed score unsuitable. Private support coefficients, common risk constraints, support-capacity investment, dynamic maintenance, and substitutes between source datasets are omitted. The general network remains unsolved. Documentation raises application value in this model; making it reduce support requirements endogenously adds another nonseparable margin and is not covered by the stated formulas.

The article establishes a joint allocation solution, a strategic scoring distinction, and an exact reserve/participation frontier for a limited accounting class. These are stronger results than the original frozen-adoption cross-partials. They remain applications of familiar incentive and club-good ingredients. The closest data-sharing comparator is Rasouli and Jordan (2021), alongside public-goods financing and internal transfer pricing. The restricted cross-type reserve calculation does not exclude better mechanisms with quality distortion, consumer assessments or richer contracts. The strongest rejection reason is that the precise frontier may be judged too dependent on common-fee and audit restrictions, and its empirical importance is unknown. The work is complete at its theoretical scope; that does not establish publication readiness or unrestricted originality.

A prospective field study is documented separately. It has no completed observations and no implemented field-trial uncertainty pipeline. Its need for partner authorization, auditable value measures, graph support, and adequate independent clusters remains unchanged. None of those future tasks is presented as an unfinished proof or a missing empirical result in this theoretical article.

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Technical appendix: complete proofs and boundaries

A. Existence, strict concavity, and the decentralized equilibrium

Write $\mathcal{Y} = \{y \geq 0 : a'y \leq \bar{L}\}$. It is nonempty and compact since $a_j > 0$ and $\bar{L} > 0$. For the unpriced program use the compact box $[0, \bar{y}]$. The block quadratic form for the support-priced program (15) has negative Hessian with positive cost matrix

$$\begin{pmatrix} \kappa K/\beta & -V' \\ -V & H \end{pmatrix}. \tag{A1}$$

Its Schur complement is $H - \beta Q/\kappa \succ 0$ by (A). Thus it is strictly positive definite for $\beta \in (0, 1]$, and the program is strictly concave and coercive in z. The same argument with H replaced by D proves strict concavity of the joint unpriced potential. Existence and uniqueness follow, including boundary optima. At zero share, $z = 0$ is uniquely cost minimizing and strict concavity in y gives a unique use allocation.

Maximizing first in y in the unpriced joint program produces (6), after multiplication by α and subtraction of the constant $\alpha G_D(0)$. This justifies the sequential comparison rather than assuming a producer ignores induced demand. On an interior box branch, the first-order conditions are $Dy = h + Vz$ and $\kappa Kz = \alpha V'y$. Eliminating z gives (7). Substituting the identities into (4) gives (8). Since $M_D = D - \alpha Q/\kappa$ is symmetric positive definite and $dy/dh_j = M_D^{-1} e_j$, differentiating the quadratic expression (8) yields (9). At entry or reservation boundaries use the unique constrained program; the interior derivative is not extrapolated across kinks. The constructed values in Table 1 directly verify a finite harmful opening across an entry boundary.

B. Full information, admission, and pricing units

For the full-information program the positive definite block form is (A1) at $\beta = 1$. Slater's condition holds for capacity using small strictly positive eligible uses. The KKT conditions are necessary and sufficient and give (10)-(11). The conditions for z are equalities because K is diagonal and $V'y \geq 0$: whenever its component is zero the associated investment is zero; otherwise the optimum is positive.

At any fixed z, a price-taking marginal user in class j requires $h_j + v_j' z - y_j/n_j - p_j = 0$ when active and a weakly nonpositive marginal gain at zero. Inserting (12) yields exactly the conditional social KKT system. Strict concavity in y guarantees uniqueness. If $\ell < \bar{L}$ then $\nu = 0$. If $\gamma = 0$ and capacity is slack then every required price is zero. With heterogeneous positive support coefficients, two active classes and $\lambda > 0$, their required application prices differ. An identical per-use price cannot satisfy both necessary interior conditions. At corners or equal support requirements the impossibility need not apply.

C. Sequential scoring and the gross-credit deviation

The optimizer defining G is unique and lies in the compact set $\mathcal{Y}$. The envelope theorem gives $\nabla G(z) = V'y(z)$, including across a change in active class where the derivative remains continuous. The producer's objective is $\beta[G(z) - G_0] - C_\kappa(z) - F$. It is equivalent to maximizing the joint program multiplied by β, because the outer maximum in z commutes with the inner maximum in y. Strict joint concavity gives a unique producer and consumer allocation. This proves (15)-(16), and using the y-KKT equations yields (17)-(18). A nonpositive proposed scarcity multiplier or an invalid inactive-class inequality means that proposed active set is discarded, not that a negative use is permitted.

On a slack-capacity positive-use region, $y(z) = H^{-1}(h + Vz)$ and $G(z) = \frac{1}{2}(h + Vz)'H^{-1}(h + Vz)$. Subtracting the actual constrained $G_0 = G(0)$ gives $G(z) - G_0 = c_0 + b'z + z'Jz/2$, proving

the level in (19). If zero investment is on the same branch or its value-preserving closure, $G_0 = \frac{1}{2}h'H^{-1}h$ and $c_0 = 0$. A baseline outside that branch can give $c_0 \neq 0$. Constants disappear only on differentiation, not from transfers or participation. Each candidate in (20) must obey its branch constraints; global comparisons use the constrained value function. The gross score is $\beta z'V'H^{-1}(h + Vz)$, which has twice the quadratic term. Differentiating gives (20). With $\beta = 1$, the correct optimum is the unique welfare optimum. If $Jz_F \neq 0$, evaluating the gross first-order condition there leaves a nonzero derivative, so the gross optimum differs; strict concavity of true welfare makes its welfare strictly smaller. The proposition is conditional on the gross objective's stated positive definiteness and interior uses; it makes no claim of a finite optimum otherwise.

D. Reserve theorem, private costs, and participation

Once the type-specific allocation is chosen, the producer's net incremental payoff is $r_\kappa - F$. Ex-post voluntary participation for every type requires $F \leq \min_\kappa r_\kappa$. Since $z = 0$ is available before the fee is paid, optimized rent is nonnegative. Conditional on acceptance, F is constant in investment and does not change its maximizer. The center's net expenditure is

$$T_\kappa + S(\ell_\kappa) + m - \lambda_\kappa \ell_\kappa - F = T_\kappa - \Gamma_\kappa + m - F. \tag{A2}$$

Requiring (A2) to be at most B for every type gives the lower bound in (24). Combining it with participation and $F \geq 0$ proves necessity and sufficiency of (23). Taking the largest participating fee minimizes the needed reserve. No cost type is reported in implementing that common fee.

If type is verifiable, choose $F_\kappa = r_\kappa$, giving zero incremental producer rent and expenditure $c_\kappa - \Gamma_\kappa + m$. This proves (25). For each type, $r_\kappa \geq \min r$, so $T_\kappa - \Gamma_\kappa - \min r \geq c_\kappa - \Gamma_\kappa$. Maximizing and taking the positive part establishes $B_I \geq B_K$. The inequality is strict in Table 2, so it is not an algebraic equality disguised as an informational constraint.

This proof establishes only what the designated fee and credit family can implement. A fee conditioned on output, a menu, or fixed consumer assessments can change both constraints. Private costs alone do not logically prohibit such mechanisms. With strict participation and a binding bound, the upper bound becomes open as stated in Section 6.

E. Monotonic welfare, existence of the constrained optimum, and audit error

Let $g_i = G(z_{\beta_i}) - G_0$ and $c_i = C_\kappa(z_{\beta_i})$ for $0 \leq \beta_1 < \beta_2 \leq 1$. Producer optimality at both shares gives

$$\beta_1(g_2 - g_1) \leq c_2 - c_1 \leq \beta_2(g_2 - g_1). \tag{A3}$$

Adding the optimality inequalities first yields $g_2 \geq g_1$. Consequently true-surplus change is $g_2 - g_1 - (c_2 - c_1) \geq (1 - \beta_2)(g_2 - g_1) \geq 0$. This argument covers entry, exit, binding capacity, and zero credit without differentiability. Constant m does not change the ordering.

Allocations and the value function are continuous in β on $[0, 1]$ under strict concavity and compact/coercive feasibility. The support multiplier is unique whenever capacity binds with positive load: an active positive class identifies it; at transitions it matches the slack value zero. At no demand choose $\nu = 0$. Thus r, T, Γ, and B_I are continuous in this model, and the feasible share set is closed in a compact interval. A nonempty set has a largest element. Applying (A3) separately to every type proves Theorem 2 for arbitrary type probabilities, without assuming the budget function is monotone. A fixed administration cost may leave the feasible set empty; the theorem explicitly conditions on nonemptiness.

On a smooth active branch differentiate $\beta \nabla G = \kappa K z$ to obtain (27). $\kappa K - \beta G_{zz}$ is positive definite by strict concavity of the reduced problem. The gradient relation shows $\kappa z' K z_\beta \geq 0$.

For a factor-inflated score, replacing β by $\widetilde{\beta}$ proves the audit-error statement whenever the effective share is at most one. Budget feasibility requires a separate worst-case calculation for actual payments. There is no claim that a failed audit bound still implements the frontier.

F. Preserved scalar benchmark and its restricted interpretation

For fixed downstream adoption, let gross reusable value per module be $B(t) \geq 0$, linear producer cost $D_0(t)$, quadratic cost curvature $K_c(t) > 0$, and private appropriation α. Then $q_D = [\alpha B - D_0]_+ / K_c$ and constrained optimum $q_C = [B - D_0]_+ / K_c$. When $B > D_0$, completing the square gives welfare loss $K_c(q_C - q_D)^2/2$. On an interior branch,

$$q_{D,\alpha} = B/K_c, \quad \widetilde{W}_\alpha = (1-\alpha)B^2/K_c. \tag{A4}$$

Differentiating in t gives quantity complementarity if $B'/B > K_c'/K_c$, and surplus complementarity if $2B'/B > K_c'/K_c$, for $\alpha < 1$. For general scalar convex costs replace K_c by cost curvature evaluated along the equilibrium branch; that curvature itself then depends on the allocation. This is the former result, fully conditional on a scalar investment object. It is not used to prove Theorems 1-2 or to infer the multi-margin signs in (28).

G. Reproduction and accounting boundaries

The active-set algorithm checks all positive-use subsets, then slack or binding capacity, using (17)-(18). For the unpriced comparison it checks lower/free/upper reservation states. It enforces inactive inequalities, positivity, and the physical limit. Independent constrained optimization uses the unreduced objective, rather than reusing the reduced first-order conditions. The exact financing certificate rewrites each requirement $T_\kappa - \Gamma_\kappa - r_{\kappa'} \leq 0$ as a rational polynomial inequality for the declared zero-reserve example, isolates all roots, and checks intervals. It is a certificate for the example, not a universal assertion that feasible shares form an interval.

No synthetic dataset is labeled as observed. Copies of external papers are not redistributed. The archive includes the original assignment, central reviews, editable sources, executable code, output tables and figure, reference audit, and numbered revision responses. The current entry point is `python3 code/reproduce_R2.py`; it reruns R1 and the exact crossing-baseline score and settlement regression in Section 8.5. The regression also verifies that the preserved numerical R1 files are byte-identical after rerunning. The historical field protocol is clearly prospective and cannot validate this mechanism until an authorized study is executed.